CivicCast User Manual

For station operators, clerks, and IT staff - v1.0.0-beta.5 (native Windows line)

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CivicCast User Manual

CivicCast is open-source, self-hostable civic meeting recording and publication software. This controlled beta proves a bounded local path: create or upload recorded media, rehearse that exact sample privately, package it privately, explicitly approve Portal publication, and play it in the resident portal. Live ingest, 24/7 station operation, hardware/headend delivery, external providers, and app stores need separate integration and field proof.

This manual is one document in three sections, each pitched at a different reader:

- **Section A — End-User Guide.** For board members, clerks' office staff, meeting operators, and anyone who is going to *use* CivicCast without needing to maintain it. No jargon. If a technical term is unavoidable, it is defined inline.
- **Section B — Technical Reference.** For the IT staff, integrators, and ops people who install, configure, monitor, and troubleshoot the station. Includes the CLI, the environment-variable surface, the credential store, and the role and permission model.
- **Section C — Architecture Reference.** For developers, auditors, and technical reviewers. Migration chain, the GStreamer playout engine, the three Protocol seams that connect the new modules to capture/publish/ alerting, the CDN-aware proxy resolver, and pointers into the v3.0 spec set.

The source inventory includes software-side PEG station-in-a-box capabilities plus control-room and virtual media studio beta work (see Comparative Capability Status in Section C). These source capabilities and their lab evidence are not stock acceptance claims and do not establish station-device, provider, app-store, or production proof. `v1.0.0-beta.2` was never published – it exists only as an internal Gate A upgrade-baseline kit. `v1.0.0-beta.4` is the current published release described in this manual, a download-only upgrade for stations already on `v1.0.0-beta.3` (CivicCast's first downloadable release, now superseded – see `docs/releases/release-truth.yaml`). `v1.0.0-beta.5` is the next candidate and the current native-Windows development candidate (an owner-held unpublished candidate) described in this manual; it does not change the beta.4 install story. It is a fresh, from-scratch native Windows product line — its version numbers do not continue from, and are not comparable to, the older `v1.0.0-rcNN` line documented for a retired WSL2-based product in a separate, private repository.

Who Reads What

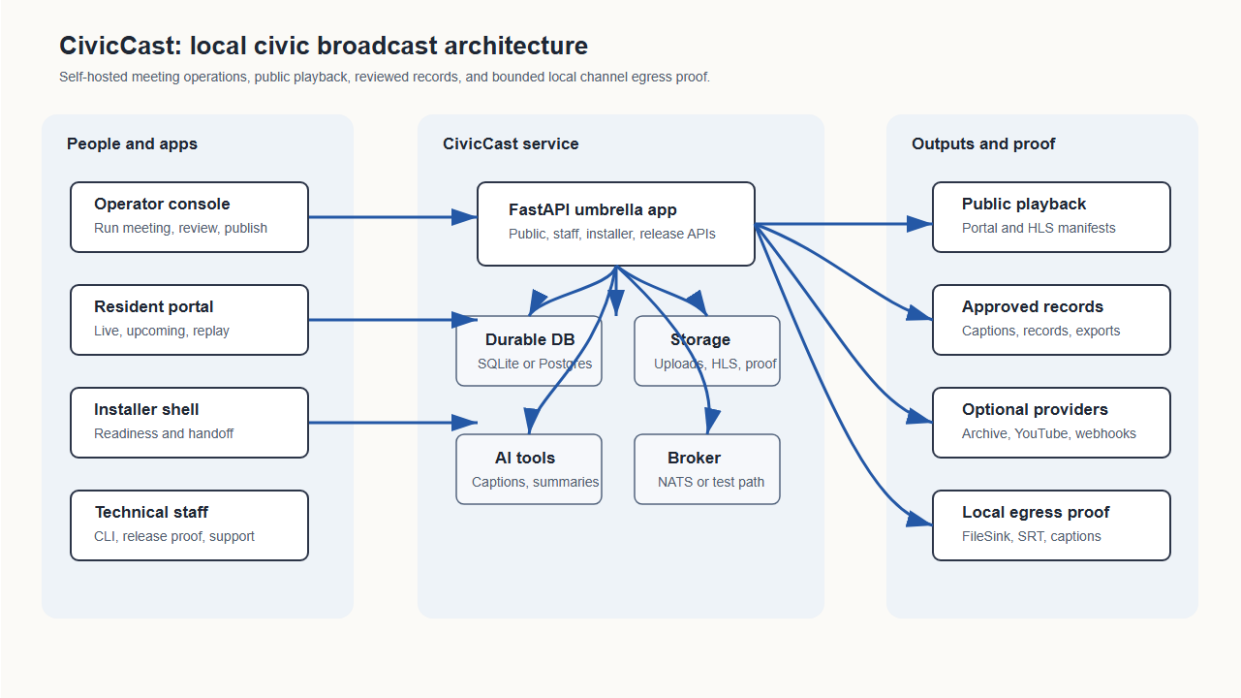

Figure 1: CivicCast system architecture

Reader	Start In	Then See
First-time user / board member / clerk	Section A	Meeting Operator Guide, Records Clerk Guide
Station admin / IT lead	Section B	Admin Guide, Technical Operations Reference
Developer / integrator / auditor	Section C	docs/spec/3.0/, RECONCILIATION.md

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Section A — End-User Guide

This section is written for people who do not live in terminals. If you can operate a meeting, post a flyer to a community board, or upload a video to a website, you can run CivicCast.

If something here points you to a screen you do not have, ask your IT lead or station admin. The page they need is in Section B — Technical Reference.

What Is CivicCast?

CivicCast's broader source inventory contains work toward the following product directions. These are not all stock acceptance claims:

- **Integrated live meetings.** With a separately proven server-side media probe (the automated check that confirms a live signal is actually present before **Start Live Stream** can turn on) and station egress (the station's own connection that sends the stream out to viewers), a council meeting can

stream to residents in real time on the public website, on the local cable channel, and (if the station turns it on) through station-operated web or OTT endpoints (apps on devices like Roku or Apple TV). Publication through commercial app stores requires provider accounts and store review and is outside this package.

- **Integrated 24/7 cable operation.** Past meetings, a community-bulletin board, and short underwriting spots can be scheduled to fill the channel between live events — without anyone in the studio.
- **Publish the record.** The source includes captions, summaries, signed PDF transcripts, and chapter markers tied to the meeting agenda. Each station must separately prove and configure the optional paths it intends to use.
- **Reach residents who do not watch live.** Subscribers get emails, a podcast feed, or app notifications when a new meeting is published.
- **Stay accountable.** Every published recording carries a tamper-evident trust stamp (an RFC 3161 timestamp). The signed PDF transcript is a review artifact; each jurisdiction decides what constitutes its legal record.

CivicCast runs on your hardware. Stock acceptance keeps the tested media local. Optional surfaces — Internet Archive mirroring, YouTube simulcast, emergency-alert ingest — turn on per-station, with the station's own accounts.

Advanced Capabilities — Roadmap, Not This Beta

To keep this manual honest about what a station can rely on today, the following are **not** ready in this beta. Each has real source code in the repository, but none has been field-proven the way the recorded-media workflow in Your First Beta Workflow has:

- **Full cable/SDI headend delivery.** The software writes the file packages and TSDuck compliance checks a cable headend needs, but no physical SDI output, QAM modulation, or PEG-headend acceptance has been proven with real hardware. See Comparative Capability Status.
- **Running more than one channel at the same time.** The engine and data model support multiple channels, but simultaneous multi-channel operation on one station has not been soak-tested the way the single-channel path has.
- **Internet Archive and YouTube syndication as a turnkey path.** Both providers have real adapters and can be configured (see Setting Up Providers, Plain Language), but each station must independently prove its own credentials and run a private upload proof before relying on either for residents — neither ships pre-verified.
- **OTT apps (Roku, Apple TV, and similar).** Native source trees exist for six targets, but publishing through commercial app stores needs the station's own developer accounts and passing that store's own review — outside this package, and not tested end to end here.
- **Emergency Alert System (EAS) hardware take-over.** CivicCast is software that can display and log CAP alerts and force a slate with explicit per-alert operator confirmation. It is **not** an EAS device, is not FCC-certified EAS equipment, and does not replace the station's existing certified EAS relay. See What if there's an emergency during a meeting?

If a board member or IT lead asks whether one of these is "done," the honest answer is: the code exists, the field proof does not yet.

Your First Beta Workflow

This walkthrough assumes someone has finished the installer and you have a username and password for the operator console.

> **Beta boundary.** The stock build has no production server-side media probe. It shows **Source preview unavailable** and keeps **Start Live Stream** disabled. That is the expected safe state, not an installation failure.

1. **Open the operator console.** Your IT lead will tell you the address (it usually looks like `https://broadcast.yourtown.gov`). Sign in.
2. **Confirm the live safety state.** Open **Run Meeting** and confirm the room says **Source preview unavailable** and does not enable live start.
3. **Create or upload sample recorded media.** Use non-sensitive test content.
4. **Run a private rehearsal.** In **System Health**, choose **Run private rehearsal**. CivicCast must report that it copied the exact validated sample, created and finalized a private recording, and loaded resident preview.
5. **Validate and package it.** An authorized publish operator or setup administrator chooses **Package for playback**. Packaging must remain private.
6. **Verify privacy before approval.** The recording must not appear or play in the resident portal yet.
7. **Approve Portal publication.** In **Publish**, select only **Portal** and approve it.
8. **Confirm resident playback.** Open the resident portal in a second browser and play the approved recording. Unapproved media must remain private.

Destructive Actions Now Confirm Before They Fire

Nearly every one-click action in the operator console that changes what residents see or takes something off air now shows a confirmation dialog before it actually runs — a second, deliberate click, not a `window.confirm` browser popup. This closed a real gap: a board member or a curious visitor standing at the console could previously end a live stream, wipe a saved configuration, or publish a recording with a single accidental click.

The dialog names the plain-language, resident-facing consequence of the action (for example, “This immediately stops sending video to residents on this channel”), and pressing **Escape** or **Cancel** does nothing — no network request is sent until you press the confirm button. Covered actions include, among others:

- **Channel Ops / Safe-to-broadcast** — Start, Stop, Restart feed, and drain, on both the Channels screen and the readiness panel.
- **Run Meeting** — End Live Stream, and Take off air.
- **Publish** — Approve and Publish selected.
- **Paywall** — regenerate the signing secret, delete the paywall config, and revoke an individual subscriber’s access.
- **Media Lifecycle** — Remove a watch-folder or retention-rule.
- **Federation (ActivityPub)** — Generate station key.
- **App Admin** — Queue build (the build form also starts with nothing selected and keeps this button

disabled until the form is actually valid).
- **Schedule** — Cancel a scheduled item.
- **Program Guide** — Disable a program slot.
- **Emergency Alert Screen** — Clear an alert, and force a slate (which already required its own separate acknowledgement checkbox before this change).
- **System Health** — Repair GStreamer runtime & restore, run a real database restore drill, run a rollback rehearsal, and open a maintenance window.

Read-only actions — checks, previews, scans, and refreshes — deliberately did **not** gain a confirmation step; a dialog on a safe action is confirmation fatigue, not safety.

Where Recordings And Backups Live

CivicCast's Windows service stores media, recordings, and backups under its own Windows account (the account the background service runs as), not under your own `C:\Users\<you>` folders. That is why "recordings are saved locally" does not mean they show up if you browse your own Documents or Desktop — the real folder is a system location a normal Windows account often cannot open directly, even by pasting the path into File Explorer.

You do not need filesystem access to find a recording. The **Assets** screen lists every recording CivicCast has, with its title, date, duration, and publish status — that is the supported way to find a recording, review it, and republish it, with no folder-browsing required.

If you do need the actual files (for a legal hold, a manual copy, or troubleshooting with IT), open **Setup → Station Profile**. The **Storage roots** section shows the exact Media library, Recordings, and Backups paths this station is using right now, each with a **Copy path** button. Paste the copied path into File Explorer's address bar; if Explorer reports access is denied, an administrator on this Windows computer can grant read access, or can run CivicCast's **Open folder** action from the station itself (it only works when you are physically at, or remoted into, the station computer — not from a different computer on the network).

The **Backup destination** field on the **Setup** screen is different: it is a folder or drive *you* choose (a local drive, an external drive, or a network share), and CivicCast verifies CivicCast can write to, read from, and clean up after itself there. A backup destination should always be a real Windows path, such as `C:\CivicCastBackups` or `D:\CivicCastBackups` — CivicCast rejects a WSL/Linux-style path (anything containing `\mnt\c\...` or `/mnt/c/...`) because that path format does not exist on a Windows station and would silently mean nothing was actually being backed up.

Managing Your Own Sign-In

Open **Setup → Station Profile** and find the **Security** panel for two account-safety actions, both available to `setup_admin`:

- **Multiple sessions are normal and supported.** Signing in on another browser or device no longer signs you out anywhere else — CivicCast keeps up to 20 signed-in sessions at once (oldest evicted first past that cap), so a laptop, a phone, and a second browser tab can all stay signed in at the same time. This is a fix from an earlier behavior where a routine sign-in elsewhere silently ended every other open session.
- **Sign out other sessions.** If a laptop or device with an open CivicCast session is lost or stolen, use

this to immediately end every *other* signed-in session while leaving the browser you're using right now signed in. It requires a second, explicit confirmation click before it fires.
- **Regenerate recovery kit.** If the one-time recovery kit from first-run setup was lost, never saved, or you just want a fresh set, this mints 8 new recovery codes and immediately invalidates every old one. It requires being signed in with the current admin password already — it is a way to replace lost codes, not a way back in if you are actually locked out.

What Each Publish Surface Means

The **Publish** dashboard shows one row per surface a recording can reach. Each row's dot color and word describe that one surface, not the whole recording:

- **Portal** (required) — the station's own resident-facing website. This is the canonical, citable public record; nothing else on the dashboard is required for a recording to be public.
- **Internet Archive**, **YouTube**, **Local archive folder**, **Subscriber notices**, **Podcast feed** — optional reach and archive surfaces (see Setting Up Providers, Plain Language). Each fans out independently on its own timeline.
- **Cable file package** — an optional local ZIP of media, captions, metadata, and hashes for handoff to a cable/PEG headend. Most stations never configure a cable-package output folder, and that is fine: this surface reads **Not set up (optional)** rather than a red failure when it was never configured. A red **Failed** on this row means CivicCast actually tried to build the package (because an output folder is configured) and hit a real problem — a missing source file or a missing caption sidecar — which the row's message names directly.

A red or unset optional surface never blocks or undoes a successful Portal publish. Check the dashboard's overall status, not any single optional row, to know whether a recording is public.

The CDN Cost Estimate Is A Guess, Not A Quote

The **Storage and viewing estimate** panel on **Setup** turns hours, meetings, and viewers into a storage figure (a straightforward multiplication) and an honest note about bandwidth cost: it does not print a single invented dollar figure, because CivicCast does not know which CDN provider a station will use or what that provider charges. Cost genuinely **varies by provider** — **Cloudflare R2 charges nothing for sending video out to viewers**, which is a real, current, published price (not a CivicCast estimate); BunnyCDN, Fastly, and Akamai each set and change their own rates, so check their pricing pages directly before budgeting a paid CDN.

Live Broadcast — What It Needs And Its Honest Limits

Live broadcast is available, but it is not turnkey the way the recorded-media workflow is. Two things to know before relying on it for a real meeting:

- **A live broadcast needs an encoder or SRT source configured first.** Without one, **Run Meeting** shows **Source preview unavailable** and keeps **Start Live Stream** disabled — that is the expected safe state described in Your First Beta Workflow, not a bug. Set up and verify the station's encoder/SRT source before the first live meeting.
- **A live-source drop mid-broadcast causes a brief re-establish, not a seamless reconnect.** If the encoder or SRT feed drops during a live broadcast, the channel does not silently keep playing the

interrupted feed and it does not instantly resume where it left off either — after a short run of failed relaunch attempts against the same dead source, CivicCast falls back to a slate (a hold screen) rather than looping the crash forever, and picks the source back up automatically once it recovers. Residents will see a brief interruption, not a seamless hand-off. Never assume the recording continued through a drop — confirm the recording and finalization status before publishing.

Operator Graphics Control (Lower-Third Banner)

Channel Ops now has a graphics-overlay panel where an operator can set a lower-third text banner (a text strip across the bottom of the picture) and turn it on or off for a channel, without editing any config file.

Honest limit: this is not a live, hot text update. A saved banner toggle or text change takes effect the next time the channel's playout pipeline builds or reloads its content — the next channel start, or the next scheduled content reload — not instantly on an already-live picture. The banner itself is a still image the engine composites into the video, not a live text-render layer, so there is no way to flash updated text onto an already-running broadcast the instant you type it. Plan a lower-third change before the meeting starts, or expect a short delay before it appears on an already-live channel.

Station bug/logo placement is not yet operator-controllable from this panel; only the lower-third text layer is.

Common Operator Questions

What happens if Wi-Fi drops mid-meeting? The stock build does not claim automatic source-drop detection or slate failover (automatically switching to a hold slide when the video source drops) without a verified media probe and station-specific egress proof (verified evidence that the station's outbound stream setup actually works). Follow the station's tested fallback procedure. Never assume the local recording continued: confirm the recording and finalization status before publishing.

Can I trim the start and end of the recording? The Assets editor is a scrub-and-metadata editor: it can save trim points and chapters, and it keeps the original. It does not claim automatic public re-rendering. Package the asset again after changing trim metadata, then review and publish it.

Where do captions come from? CivicCast generates captions automatically from the meeting audio using a local AI model — your audio never leaves the station unless you turn on a cloud model in settings. A records clerk reviews and corrects the captions before the recording is published.

What if my station's captions quietly drop to a lower-quality model? If the station's large caption model can't be verified at startup (for example, after an upgrade), CivicCast automatically falls back to its proven standard-tier model rather than failing to start — but it no longer does this silently. **System Health** raises a visible **"Captions are running on the standard tier … open AI Models"** alert so staff know to re-check the model, instead of running degraded captions with no on-screen sign anything changed.

Why does an asset's readiness dot say "Not ready" even though it's already published? The small readiness dot on **Assets** tracks the optimized playback proxy the ingest pipeline builds, not whether the recording is public. A published, live-on-the-portal asset can still show a not-ready proxy dot — hover it (or check the screen-reader text) for the plain-language reason, and check the separate **Published** column for actual publish status.

I can't find the upload button on Assets. It's there: **Assets** has its own **Upload video** button (separate

from the First Setup rehearsal picker), with a progress bar and a plain-language error if the file type isn't supported. It's gated to `records_clerk`, `meeting_operator`, and `support_admin` — if your role lacks it, the button is visibly disabled with the reason stated, not hidden.

A watch-folder file didn't ingest — how do I know why? Each watch folder now shows its own health status, last poll time, last ingest time, and — when something's wrong (an unreachable path, a permission problem) — the actual degraded reason, instead of failing silently. Use **Scan now** on the folder's card to force an immediate check instead of waiting for the next automatic poll.

How do residents find a recording? The public website has a **Browse** page with search and filters. Each recording also has a shareable link with a copy button. If your station tags recordings by meeting body (e.g. *City Council*, *Planning Commission*), residents can filter by that.

Do I need to upload anything to YouTube or Apple Podcasts myself? Optional providers require station credentials, configuration, and separate controlled proof. The clean-install walkthrough proves Portal-only publication; it does not claim that YouTube, Internet Archive, podcast directories, or other external providers are ready for a station.

How does a recording become public? In **Assets**, package the validated recording. Packaging prepares private HLS media but does not publish it. In **Publish**, approve the **Portal** surface. Only a successful Portal approval makes the recording metadata and HLS media resident-visible.

What does the restore rehearsal prove? The System Health action runs a real, isolated database restore drill. In the clean-host walkthrough it verified 95 tables and crash recovery. It does not prove recovery of media, configuration, or credentials; those require separate station backup and restore procedures.

How do I send support a bundle from Windows? In **System Health**, add a short note, choose **Create support bundle**, then choose **Download support bundle**. Send the downloaded JSON file and its displayed SHA-256 through the private beta support channel.

What if there's an emergency during a meeting? CivicCast contains optional EAS (Emergency Alert System) / CAP (Common Alerting Protocol) software surfaces, but CivicCast does not claim a hardware-certified EAS relay or station take-over path. Use the station's existing certified emergency procedure unless that integration has separate field proof. If EAS is not configured and proven, CivicCast does not interrupt the broadcast.

The meeting is closed-door — can I still record it without broadcasting? Yes. On **Run Meeting**, choose **Record only** instead of **Live**. The recording is stored privately until staff publish it.

Setting Up Providers, Plain Language

Setup has a **Provider setup** card for each optional service CivicCast can use. Every card is either **Required** (needed for the local tester path) or **Optional** (the station can skip it and everything else keeps working). Only **Local resident portal** and **Backup destination** are required. Every provider below is optional — read its card, decide if the station wants it, and skip it otherwise. Nothing else on this list blocks a broadcast.

You do not need a "technical admin" for any of these. If you can create a free account on a website and paste a code into a box, you can set these up yourself. Each card also has a **Setup guide** you can open for the exact steps, and a link back to the matching part of this manual.

Cloudflare R2 (recommended, usually free) Cloudflare R2 is CDN storage (see Glossary) that CivicCast can set up for you automatically. On the **Cloudflare R2** card, use the **CDN concierge** box: create a free Cloudflare account, create one API token, paste it in, and click **Provision for me**. CivicCast creates the bucket and turns on public access for you — you never have to know what a bucket, an object store, or a CDN pull-zone is. Cloudflare R2 does not charge for sending video out to viewers ("egress"), which is why it is CivicCast's recommended first choice. If you would rather enter the account ID, access key, secret key, bucket, and public URL by hand, the same card has a manual fallback below the concierge box.

Internet Archive (optional, permanent public copy) Internet Archive keeps a second, independent public copy of every published recording, hosted by a nonprofit library and outside CivicCast's control — useful if the station wants an archival copy that survives even if the station's own server ever goes away. To set it up: create a free account at archive.org, open its **S3 keys** page (Internet Archive names its keys after Amazon S3 because it uses the same style of key, not because Amazon is involved), copy the access key and secret key it shows you, and paste both into the Internet Archive card. There is no "technical admin" step — paste your own keys in yourself.

YouTube (optional, extra reach) Turn this on only if the station wants an extra copy of meetings on its own YouTube channel. You will need a Google **OAuth client ID and secret** — a matched pair of codes, created for free at Google Cloud Console, that let CivicCast upload to the channel without ever seeing the channel's password. Create them, paste both into the YouTube card, and run a private upload proof before using YouTube for residents.

Subscriber notices (optional) Turn this on to email or webhook-notify subscribers when a new meeting is published. A **webhook secret** is a password-like string shared between CivicCast and the notification service, so each side can prove a notification really came from the other — CivicCast can generate one for you, or you can paste one from your notification provider.

Local archive folder (optional, second local copy) Point this at a second drive, external drive, or network share if the station wants an extra local copy beyond the built-in Recordings folder (see Where Recordings And Backups Live). Enter the folder path; CivicCast proves it can write to, read from, and clean up after itself in that folder before marking this ready.

BunnyCDN, Fastly, and Akamai (optional, alternative CDNs) These are alternative CDN/object-storage providers for a station that already has an account with one of them, or wants a paid alternative to Cloudflare R2. Unlike R2, these providers typically charge for sending video out to viewers — check their own pricing pages before choosing one; CivicCast does not know or estimate what they will charge (see The CDN Cost Estimate Is A Guess, Not A Quote). BunnyCDN calls its address for viewers a **pull-zone** (something like `your-zone.b-cdn.net`) and its file storage a **storage zone**; Fastly and Akamai use the more generic **bucket** and **region** terms from the glossary. All three cards need an account, a storage/object area, an access key, and (for Fastly/Akamai) a region — the card's Setup guide lists the exact fields.

Federation / ActivityPub (optional, advanced, off by default) Federation lets other services that speak the ActivityPub protocol — the network behind Mastodon and similar sites, sometimes called "the fediverse"

— follow the station and see when a new meeting is published, the same way someone might follow a page on a social network. Most stations do not need this and can leave it off. If the station wants it, open **ActivityPub** in the console and choose **Generate station key** — CivicCast creates the station's federation identity for you; no command line is required. See the station's ActivityPub screen for the current on/off switch and approval queue.

Podcast feed (optional) The podcast feed republishes a meeting's audio as a podcast episode once that meeting has already gone through the normal flow: recorded, packaged, and approved on the **Portal** publish surface (see What Each Publish Surface Means). "Publish a local portal recording first" on this card's Setup guide means exactly that — there is nothing extra to configure before the first Portal-published recording appears here for review.

Glossary

Plain-language definitions for the technical terms that show up on provider setup cards. If a term you need is not here, its provider's card also links back to this section.

Term	What it actually means
S3 access key / secret key	A username-and-password-style pair (the "access key" is the username, the "secret key" is the password) that proves to a storage provider it is really the station's account making a request. "S3" was Amazon's name for this style of storage API; other providers (Internet Archive, Cloudflare R2, BunnyCDN, Fastly, Akamai) reuse the same key style even though they are not Amazon.
Object store / bucket	A place to store files in the cloud, organized differently from a Windows folder. A "bucket" is the named container inside that store — think of it as the top-level folder CivicCast's videos live in once they leave the station.
CDN (content delivery network)	A network of computers around the world that keeps a copy of the station's video near each viewer, so a meeting with hundreds of viewers loads quickly instead of overloading the station's own internet connection.
Pull-zone	BunnyCDN's name for the web address a CDN gives a station's video once it is set up (for example `your-zone.b-cdn.net`). Other CDNs call the same idea a "distribution" or "custom domain."

Term	What it actually means
OAuth client ID / client secret	A pair of codes a service (like Google/YouTube) issues so CivicCast can act on the station's behalf — for example, upload videos — without CivicCast ever seeing or storing the station's actual account password.
Webhook secret	A shared password-like string that lets CivicCast and another service each prove a notification message really came from the other, instead of from someone pretending to be them.
Egress	The data sent **out** to viewers when they watch a video, as opposed to the data **stored**. Most CDN and storage providers charge separately for egress; Cloudflare R2 is the one CivicCast recommends first because it charges nothing for it.

When to Ask for Help

Send this to your IT lead, station admin, or open a GitHub issue at the project's repository:

- The "Run Meeting" page says **Do not broadcast yet** and the yellow notice points to a step you do not have access to.
- The live stream shows on the operator console but residents see a blank or "stream offline" page on the public website.
- Captions are blank, garbled, or stuck on one line.
- A recording is missing from the **Assets** list after a meeting.
- CivicCast asks for a recovery code and the kit cannot be found.

Don't Have A GitHub Account?

Every **Report a beta issue** link opens a GitHub bug template, which needs a free GitHub account to submit. If you do not have one and do not want to make one:

1. In **System Health**, add a short note describing what happened, then choose **Create support bundle** and **Download support bundle**. This saves one redacted JSON file with the station's version, setup state, and recent activity — CivicCast strips passwords, tokens, private keys, and subscriber data from it automatically before it is saved.
2. Ask your IT lead, station admin, or anyone else at the station with a GitHub account to paste your description and attach the downloaded file to a new issue at the project's repository.
3. If nobody at the station has a GitHub account, email the file and a short description straight to the project maintainer at the address listed in SECURITY.md (that address exists for exactly this: reports from people who cannot use GitHub). Never post passwords, recovery codes, staff tokens, or private meeting material anywhere, including in an email.

For routine workflow questions, use the role-specific guides:

Admin Quick Guide

- Admin Guide — first install, recovery, backups.

Meeting Operator Quick Guide

- Meeting Operator Guide — night-of-meeting.

Records Clerk Quick Guide

- Records Clerk Guide — caption review, publish.
- Operator Language Guide — the words the console uses and what they mean.

For a problem the role guides do not cover, the project's GitHub Issues page is the right place. The Windows installer, operator console, and resident portal each include a **Report a beta issue** link that opens the project's bug template. Do not paste passwords, recovery codes, staff tokens, subscriber lists, private meeting material, or unedited meeting recordings into a bug report.

Section B — Technical Reference

This section is for the IT staff, integrators, and operators who maintain the station. It assumes a working knowledge of the shell, environment variables, and HTTP services.

For deeper command, credential, certificate, storage, model, and release-proof detail, see Technical Operations Reference. For the narrative install procedure on Windows, see INSTALL-WINDOWS.md.

Install And First Boot

The source supports two installation paths. Windows beta use is limited to the bounded recorded-media workflow described in Section A.

1. **Windows installer (controlled beta for testing).** The release `.exe` from the GitHub Release page registers a Windows service through the SCM, which supervises the control plane, Postgres, and the media workers from a bundled runtime under `C:\Program Files\CivicCast (Native)\`, and walks the operator through a recovery-kit and first-admin flow. Verify the SHA-256 hash against the release's `.sidecar.json` before running it. Do not assume the candidate is signed: compare its actual Authenticode status and publisher with the exact approved handoff. See INSTALL-WINDOWS.md.
   Leave at least **5 GB free** for the base installation. Recordings, station media, backups, and downloaded caption models require additional storage.
   Local AI models. The local AI models (Ollama summary and translation models, roughly 15-20 GB combined) are large; CivicCast ensures the same three-tag target set and downloads only the tags still missing, automatically in the background after the base install finishes, not before, and a slow or failed model download does not block the operator console from opening.
2. **Source / uv (for developers and integrators).**
   ```
   uv sync --all-extras --group dev
   uv run civiccast --version
   uv run civiccast doctor
   uv run uvicorn civiccast.app:app --reload
   ```

Without `DATABASE_URL`, CivicCast starts in local setup mode and prepares installer-managed durable SQLite storage. Set `DATABASE_URL` to use Postgres in production.

After install, open the operator console at `http://localhost:8000/operator/` (or the installer-provided operator handoff URL), confirm **System Health** is green, save the recovery kit, and run a private first-broadcast rehearsal before the first public meeting.

Updating To A New Version

Use the complete new CivicCast kit and run its installer directly on the station that already has CivicCast (Native) installed. You do **not** need to uninstall the current version first.

1. Before the maintenance window, save a current recovery kit and confirm your normal station backup is available.
2. Stop active meetings, recordings, publishing jobs, and other operator work. Close the CivicCast desktop window.
3. Keep the new `setup.exe`, its `packs` folder, and its `station` folder together, then run `setup.exe` as an administrator.
4. Setup detects the existing install and asks the old CivicCast bootstrap to stop and unregister its native service state before replacing application files. It preserves `C:\ProgramData\CivicCast`, including recordings, database data, and station settings. The new version then adopts that data and migrates its database schema.
5. Confirm **System Health** is green and spot-check a known recording in **Assets** before resuming station use.

If setup cannot prove that the old service was safely stopped, it exits with an error **before replacing application files**. Do not delete `C:\ProgramData\CivicCast`; retain the installer log at `C:\ProgramData\CivicCast\install-progress.log` and resolve the reported service error before retrying.

The Windows uninstall entry remains available for intentionally removing the application. Uninstall removes application/runtime files but deliberately leaves recordings, database data, and station settings in `C:\ProgramData\CivicCast`. Its optional checkbox concerns the signed-in account's saved installer preferences; it is not a data-purge control.

Roles And Permissions

CivicCast has exactly five roles, defined in `civiccast/auth/roles.py`. Every operator-facing endpoint requires one of them.

Role	Purpose	Owns
`setup_admin`	First-install identity. Configures the station, recovery, storage, credentials, providers, and channel settings.	Setup wizard, channel automation, credentials, provider toggles, recovery codes.
`meeting_operator`	The person running tonight's meeting.	Run Meeting, preflight, start/stop broadcast, live captions tap.

Role	Purpose	Owns
`records_clerk`	Reviews and publishes the public record.	Caption review, summary review, signed-record export, publish approvals, chapter editing.
`publish_operator`	Day-to-day program-guide and cable-channel staff.	Program guide, schedule, bulletins, underwriting trafficking, asset metadata.
`support_admin`	Diagnostics and support. Can collect a redacted bundle and see system health, but cannot change records or publish.	Support bundles, system-health detail, logs.

Role aliases (kebab-case and dotted) are accepted by the API but canonical-cased forms are the ones written to the database. A single staff member can hold multiple roles.

Staff bearer tokens remain valid until revoked or rotated. Lifecycle-token secrets are shown once; CivicCast stores a salted PBKDF2 verifier plus a full-secret SHA-256 fingerprint used only for exact admission after a peer's failure budget is saturated. Legacy environment tokens remain valid while their configuration remains active only when they use the current versioned format from `civiccast token generate-env`. Headless callers (CI, the egress worker, cable packaging scripts) authenticate via `CIVICCAST_STAFF_TOKENS` or the per-token store managed through `civiccast token ...` subcommands.

Environment Variables

The exhaustive list of `CIVICCAST_*` variables CivicCast reads is below. Defaults are documented in the source comment of the consuming module. None of these are required for a default Windows installer flow — the installer writes a configuration file with the defaults that a typical station needs.

Identity, storage, and first-run

- **`CIVICCAST_STATION_ID`** — Short station identifier used in evidence files and headend handoff.
- **`CIVICCAST_STATION_TZ`** — IANA timezone (e.g. `America/New_York`) for schedules and as-run reports. Optional override: normally the service reads the timezone the operator chose during first-admin setup (persisted in the local station-state file); set this only to force a different zone without re-running setup.
- **`CIVICCAST_ROOT`** — Override for the durable-data root. Default: the installer-managed path.
- **`CIVICCAST_CONFIG_DIR`** — Config-file root. Default: under `CIVICCAST_ROOT`.
- **`CIVICCAST_MANAGED_STORAGE_DIR`** — Where SQLite durable storage lives when no `DATABASE_URL` is set.
- **`CIVICCAST_BACKUP_DIR`** — Managed backup target used by installer backup verification and disaster-recovery workflows.
- **`CIVICCAST_UPLOAD_DIR`** — Operator and contributor upload landing.
- **`CIVICCAST_UPLOAD_MAX_BYTES`** — Per-upload cap.
- **`CIVICCAST_EVIDENCE_DIR`** — Where proof artifacts (continuity proof, soak logs) are written.

- **`CIVICCAST_VERSION`** — Reported and update-checked version pins.
- **`CIVICCAST_AVAILABLE_VERSION`** — Reported and update-checked version pins.
- **`CIVICCAST_ALLOW_FIRST_ADMIN_RESET`** — Permit a setup_admin reset after first-admin loss.
- **`CIVICCAST_ALLOW_EPHEMERAL_STORES`** — Permit in-memory stores. **Tests only.**
- **`CIVICCAST_ALLOW_INSECURE_MANIFEST`** — Permit an unsigned install manifest. **Tests only.**

Staff identity and tokens

- **`CIVICCAST_STAFF_TOKENS`** — Semicolon-separated `token:operator_id:display_name:role[,role...]` entries for explicit headless or recovery compatibility. Roles are required; generate each versioned random bearer secret with `civiccast token generate-env`. Prefer lifecycle tokens for routine operation.
- **`CIVICCAST_STAFF_TOKENS_FALLBACK_WITH_DB`** — Allow the env-var tokens to coexist with DB-issued ones.
- **`CIVICCAST_ALLOW_DETERMINISTIC_STAFF_TOKEN`** — Use deterministic or short fixture tokens. **Tests only; never enable at a station.**
- **`CIVICCAST_AUTH_ACK`** — Acknowledgement gate for the auth subsystem on first run.
- **`CIVICCAST_AUTH_RATE_LIMIT`** — Failed-authentication budget. Default: `10`. Setup uses peer-plus-route keys; staff bearer auth uses one observed-peer key across all staff routes. Valid staff tokens do not count.
- **`CIVICCAST_AUTH_RATE_LIMIT_WINDOW_SECONDS`** — Sliding failure-window duration in seconds. Default: `60`.
- **`CIVICCAST_REAL_BOUNDARY_TOKEN_FILE`** — File holding the "real provider" enable token (prevents accidental live calls).

Channel egress, GStreamer engine, automation

- **`CIVICCAST_EGRESS_ENGINE`** — `gstreamer` (default, S15) or `ffmpeg-concat` (legacy fallback).
- **`CIVICCAST_EGRESS_WORK_DIR`** — Per-channel temporary directory for egress plans.
- **`CIVICCAST_EGRESS_EMBED_CAPTIONS`** — Attempts the native GStreamer caption-SEI path when the required GStreamer elements are available. The Windows installer stages the CivicCast-bundled private GStreamer runtime under the install root's `runtime\dependencies\gstreamer` and verifies `cccombiner`, `ccconverter`, `h264ccinserter`, and `tttocea608` with the bundled `gst-inspect-1.0.exe`; if any required element remains unavailable, setup stops with an explicit native caption-SEI runtime error instead of claiming the embed path is ready.
- **`CIVICCAST_GST_ALLOW_HARDWARE_DECODE`** — Optional expert override. By default, the GStreamer worker demotes GPU H.264/H.265 decoders so live UDP/SRT decode stays in system memory before the CPU conform/encode chain. Set to `1` only when the station has validated its hardware decode path end-to-end.
- **`CIVICCAST_CHANNEL_AUTOMATION`** — Enable the channel automation driver and its poll cadence.
- **`CIVICCAST_CHANNEL_AUTOMATION_POLL_SECONDS`** — Enable the channel automation driver and its poll cadence.
- **`CIVICCAST_AUTOSCHEDULE`** — Enable the S19 query-driven autoscheduler worker.
- **`CIVICCAST_PROGRAM_LOG_WORKER`** — Program-log materializer cadence.

- **CIVICCAST_PROGRAM_LOG_POLL_SECONDS** — Program-log materializer cadence.
- **CIVICCAST_PROGRAM_LOG_HORIZON_HOURS** — Program-log materializer cadence.
- **CIVICCAST_FINALIZATION_WORKER** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_POLL_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_MAX_ATTEMPTS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_BACKOFF_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_SETTLE_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_NEVER_APPEARED_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_RUNNING_LEASE_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_RETENTION_WORKER** — Retention worker controls.
- **CIVICCAST_RETENTION_POLL_SECONDS** — Retention worker controls.
- **CIVICCAST_BULLETIN_EXPIRY** — CG bulletin expiry worker.
- **CIVICCAST_RELOAD_TIMEOUT_S** — Worker-supervision timeouts.
- **CIVICCAST_STALL_TIMEOUT_S** — Worker-supervision timeouts.

Captions, EAS, alerting

- **CIVICCAST_CAPTION_TAP** — Live caption tap configuration.
- **CIVICCAST_CAPTION_TAP_DIR** — Live caption tap configuration.
- **CIVICCAST_CAPTION_TAP_POLL_SECONDS** — Live caption tap configuration.
- **CIVICCAST_CAPTION_TAP_SEGMENT_SECONDS** — Live caption tap configuration.
- **CIVICCAST_CAPTION_FEED_POLL_SECONDS** — Caption feed and decode-back proof cadence.
- **CIVICCAST_CAPTION_PROOF_POLL_SECONDS** — Caption feed and decode-back proof cadence.
- **CIVICCAST_EAS** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_EAS_AUTO_SURFACE** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_EAS_POLL_SECONDS** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_ALERTING** — Master switch for the operational alerting service.
- **CIVICCAST_ALERT_CREDENTIALS_FILE** — JSON file holding Twilio SMS credentials.
- **CIVICCAST_EVENTS** — Event-bus on/off.

CDN, public base URL, trusted proxies

- **CIVICCAST_PUBLIC_BASE_URL** — The URL residents see (`https://broadcast.town.gov`).
- **CIVICCAST_LIVE_MANIFEST_BASE_URL** — Override for the live-stream manifest base.
- **CIVICCAST_CDN_PROVIDER** — `bunny`, `cloudflare`, or `cloudflare_r2`. Wires the trusted-proxy resolver to the provider's edge CIDRs.
- **CIVICCAST_CDN_STUB_ROOT** — Local stub directory for CDN dry-run testing.
- **CIVICCAST_TRUSTED_PROXY_CIDRS** — Explicit comma-separated CIDR list, unioned with the provider preset.
- **CIVICCAST_TRUST_PRIVATE_PROXIES** — Default `false`. When `false`, an `X-Forwarded-For` header is honored only from a proxy in `CIVICCAST_TRUSTED_PROXY_CIDRS`, so a direct caller cannot spoof its source address. Set `true` only when the station genuinely sits behind a trusted

RFC1918/loopback reverse proxy that sets the header itself.

- **`CIVICCAST_BUNNY_STORAGE_ZONE`** — BunnyCDN credentials and hostname.
- **`CIVICCAST_BUNNY_ACCESS_KEY`** — BunnyCDN credentials and hostname.
- **`CIVICCAST_BUNNY_CDN_HOSTNAME`** — BunnyCDN credentials and hostname.
- **`CIVICCAST_R2_ACCOUNT_ID`** — Cloudflare R2 origin credentials.
- **`CIVICCAST_R2_ACCESS_KEY_ID`** — Cloudflare R2 origin credentials.
- **`CIVICCAST_R2_SECRET_ACCESS_KEY`** — Cloudflare R2 origin credentials.
- **`CIVICCAST_R2_BUCKET`** — Cloudflare R2 origin credentials.
- **`CIVICCAST_R2_PUBLIC_BASE_URL`** — Cloudflare R2 origin credentials.

TSA / records / RFC 3161

- **`CIVICCAST_TSA_ENABLE`** — Enable RFC 3161 timestamp signing on signed records.
- **`CIVICCAST_TSA_URL`** — TSA endpoint URL. Default: FreeTSA (`https://freetsa.org/tsr`).
- **`CIVICCAST_TSA_POLICY_OID`** — Optional policy OID to assert in the TSA request.
- **`CIVICCAST_VERAPDF_ARTIFACT_DIR`** — veraPDF artifact directory for the signed-record export proof.

Providers (publish, archive, syndicate) These are set per kind. The default for every kind is the deterministic mock; set the per-kind variable to `real` and supply credentials to use the real adapter.

- **`CIVICCAST_PROVIDER_INTERNET_ARCHIVE`** — `mock` (default) or `real`.
- **`CIVICCAST_PROVIDER_YOUTUBE`** — `mock` (default) or `real`.
- **`CIVICCAST_PROVIDER_MAIL`** — `mock` (default) or `real`.
- **`CIVICCAST_PROVIDER_WEBHOOK`** — `mock` (default) or `real`.
- **`CIVICCAST_PROVIDER_LOCAL_NAS`** — `mock` (default) or `real`.
- **`CIVICCAST_PROVIDER_API_KEY`** — Provider credential bundle.
- **`CIVICCAST_PROVIDER_CREDENTIALS_FILE`** — Provider credential bundle.
- **`CIVICCAST_PROVIDER_PROOFS_FILE`** — Provider credential bundle.
- **`CIVICCAST_IA_ACCESS_KEY`** — Internet Archive S3-like credentials.
- **`CIVICCAST_IA_SECRET_KEY`** — Internet Archive S3-like credentials.
- **`CIVICCAST_IA_COLLECTION`** — Internet Archive S3-like credentials.
- **`CIVICCAST_IA_ITEM_PREFIX`** — Internet Archive S3-like credentials.
- **`CIVICCAST_IA_ACCESS_VALUE`** — Internet Archive S3-like credentials.
- **`CIVICCAST_INTERNET_ARCHIVE_ACCESS_KEY`** — Internet Archive S3-like credentials.
- **`CIVICCAST_YOUTUBE_CLIENT_ID`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_YOUTUBE_CLIENT_SECRET`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_YOUTUBE_REFRESH_TOKEN`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_YOUTUBE_REFRESH_VALUE`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_YOUTUBE_PRIVACY`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_YOUTUBE_MEDIA_ROOT`** — YouTube OAuth credentials and upload defaults.
- **`CIVICCAST_SMTP_HOST`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_SMTP_PORT`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_SMTP_USERNAME`** — SMTP for the email subscriber adapter.

- **`CIVICCAST_SMTP_PASSWORD`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_SMTP_STARTTLS`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_SMTP_FROM`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_EMAIL_FROM`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_EMAIL_SMTP_URL`** — SMTP for the email subscriber adapter.
- **`CIVICCAST_NAS_TARGET`** — Local NAS archive target.
- **`CIVICCAST_NAS_ARCHIVE_PATH`** — Local NAS archive target.

Subscribe (email + webhook)

- **`CIVICCAST_SUBSCRIBE_TOKEN_SECRET`**
  HMAC secret for magic-link confirmation tokens.
- **`CIVICCAST_SUBSCRIBE_ENCRYPTION_KEY`**
  Encryption key for subscriber records at rest.
- **`CIVICCAST_SUBSCRIBE_LEGACY_ENCRYPTION_KEYS`**
  Rotation helpers for older secrets.
- **`CIVICCAST_SUBSCRIBE_LEGACY_TOKEN_SECRETS`**
  Rotation helpers for older secrets.
- **`CIVICCAST_SUBSCRIBE_ACCEPT_V08_LEGACY_SECRETS`**
  Rotation helpers for older secrets.
- **`CIVICCAST_SUBSCRIBE_SECRETS_FILE`**
  File-backed alternative to the env-var secret.
- **`CIVICCAST_SUBSCRIBE_RATE_LIMIT`**
  Subscribe form rate limiting.
- **`CIVICCAST_SUBSCRIBE_RATE_LIMIT_WINDOW_SECONDS`**
  Subscribe form rate limiting.
- **`CIVICCAST_SUBSCRIBER_WEBHOOK_SECRET`**
  Webhook signature secret.
- **`CIVICCAST_WEBHOOK_SIGNING_VALUE`**
  Webhook signature secret.
- **`CIVICCAST_WEBHOOK_RETRY_WORKER`**
  Webhook retry worker controls.
- **`CIVICCAST_WEBHOOK_RETRY_BACKOFF_SECONDS`**
  Webhook retry worker controls.
- **`CIVICCAST_WEBHOOK_RETRY_MAX_ATTEMPTS`**
  Webhook retry worker controls.
- **`CIVICCAST_WEBHOOK_RETRY_POLL_SECONDS`**
  Webhook retry worker controls.
- **`CIVICCAST_WEBHOOK_TIMEOUT_SECONDS`**
  Webhook retry worker controls.

`CIVICCAST_ALLOW_DETERMINISTIC_SUBSCRIBE_SECRETS` enables deterministic test secrets. It is for tests only and must never be enabled at a station.

Analytics

- **`CIVICCAST_PUBLIC_ANALYTICS_KEY`** — Public key the resident-portal beacon uses.
- **`CIVICCAST_PUBLIC_ANALYTICS_ALLOWED_ORIGINS`** — Analytics ingestion controls.
- **`CIVICCAST_PUBLIC_ANALYTICS_RATE_LIMIT_PER_MINUTE`** — Analytics ingestion controls.
- **`CIVICCAST_TRUSTED_ANALYTICS_RATE_LIMIT_PER_MINUTE`** — Analytics ingestion controls.
- **`CIVICCAST_ANALYTICS_RATE_LIMIT_MAX_BUCKETS`** — Analytics ingestion controls.
- **`CIVICCAST_ANALYTICS_RETENTION_DAYS`** — Days to keep raw analytics events.
- **`CIVICCAST_ANALYTICS_TRUSTED_PROXY_CIDRS`** — Analytics-specific trusted-proxy CIDRs.

ActivityPub federation

- **`CIVICCAST_ACTIVITYPUB_MODE`** — `off`, `lab`, or `live`.
- **`CIVICCAST_ACTIVITYPUB_BASE_URL`** — Identity.
- **`CIVICCAST_ACTIVITYPUB_HANDLE`** — Identity.
- **`CIVICCAST_ACTIVITYPUB_DISPLAY_NAME`** — Identity.
- **`CIVICCAST_ACTIVITYPUB_PRIVATE_KEY_PATH`** — HTTP-signature keypair.
- **`CIVICCAST_ACTIVITYPUB_PUBLIC_KEY_PEM`** — HTTP-signature keypair.
- **`CIVICCAST_ACTIVITYPUB_ALLOWLIST`** — Federation domain filters.
- **`CIVICCAST_ACTIVITYPUB_ALLOWLIST_FILE`** — Federation domain filters.
- **`CIVICCAST_ACTIVITYPUB_BLOCKLIST`** — Federation domain filters.
- **`CIVICCAST_ACTIVITYPUB_BLOCKLIST_FILE`** — Federation domain filters.
- **`CIVICCAST_ACTIVITYPUB_AUTHORIZED_FETCH`** — Strict fetch mode and lab-loopback override.
- **`CIVICCAST_ACTIVITYPUB_LAB_ALLOW_LOCAL`** — Strict fetch mode and lab-loopback override.
- **`CIVICCAST_ACTIVITYPUB_INBOX_RATE_LIMIT`** — Retry and rate-limit controls.
- **`CIVICCAST_ACTIVITYPUB_INBOX_RATE_WINDOW_SECONDS`** — Retry and rate-limit controls.
- **`CIVICCAST_ACTIVITYPUB_RETRY_WORKER`** — Retry and rate-limit controls.
- **`CIVICCAST_ACTIVITYPUB_RETRY_BACKOFF_SECONDS`** — Retry and rate-limit controls.
- **`CIVICCAST_ACTIVITYPUB_RETRY_MAX_ATTEMPTS`** — Retry and rate-limit controls.
- **`CIVICCAST_ACTIVITYPUB_RETRY_POLL_SECONDS`** — Retry and rate-limit controls.

Headend handoff: NDI, SDI, TSDuck

- **`CIVICCAST_NDI_RELAY`** — NDI output relay configuration.
- **`CIVICCAST_NDI_SENDER`** — NDI output relay configuration.
- **`CIVICCAST_NDI_FFMPEG`** — NDI output relay configuration.
- **`CIVICCAST_NDI_RUNTIME_DIR`** — NDI output relay configuration.
- **`CIVICCAST_SDI_RELAY`** — SDI output relay (descoped from 3.0 default but available).
- **`CIVICCAST_SDI_FFMPEG`** — SDI output relay (descoped from 3.0 default but available).
- **`CIVICCAST_TSDUCK_HOME`** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **`CIVICCAST_TSDUCK_PATH`** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a

pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.

- **`CIVICCAST_TSDUCK_RCVBUF_BYTES`** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **`CIVICCAST_TSDUCK_NETWORK_TESTS`** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **`CIVICCAST_CABLE_PACKAGE_OUTPUT_DIR`** — Cable file-package output.
- **`CIVICCAST_CABLE_CAPTIONS_DIR`** — Cable file-package output.

Remote contribution & production control room

- **`CIVICCAST_REMOTE_CONTRIBUTION`**
  S17 VDO.Ninja guest-input service.
- **`CIVICCAST_REMOTE_CONTRIBUTION_HOME`**
  S17 VDO.Ninja guest-input service.
- **`CIVICCAST_REMOTE_CONTRIBUTION_POLL`**
  S17 VDO.Ninja guest-input service.
- **`CIVICCAST_REMOTE_CONTRIBUTION_VDO_URL`**
  S17 VDO.Ninja guest-input service.
- **`CIVICCAST_CONTRIBUTOR_STORE_PATH`**
  Contributor record store and upload landing.
- **`CIVICCAST_CONTRIBUTOR_UPLOAD_DIR`**
  Contributor record store and upload landing.
- **`CIVICCAST_TURN_HOST`** / **`CIVICCAST_TURN_PORT`**
  The TURN server guests connect through to punch out from behind NAT. coturn has no native Windows build, so on Windows CivicCast does **not** run a local coturn process: point these two variables at a **documented external TURN server** — a coturn instance on a separate Linux/BSD host, or a managed TURN provider — and leave `CIVICCAST_COTURN_COMMAND` unset. The operator console's Remote Contribution screen (Diagnostics drawer, support_admin role) shows the currently-configured host/port, a **Test TURN connectivity** button that probes it right now, and the same guidance text under "How to point this station at coturn." On Linux/macOS these still work the same way if you'd rather point at an external server than run coturn locally.
- **`CIVICCAST_COTURN_COMMAND`**
  Linux/macOS only — the local coturn launch command, if you're running coturn on the station itself rather than pointing at an external server (see `CIVICCAST_TURN_HOST` above). Leave unset on Windows; there is no native Windows build.
- **`CIVICCAST_VDO_COMMAND`**
  TURN/VDO co-process commands.

- **`CIVICCAST_CONTROL_ROOM_TSR_URL`**
  S16 production-control-room TSR endpoint.
- **`CIVICCAST_TSR_PORT`**
  S16 production-control-room TSR endpoint.
- **`CIVICCAST_TSR_NO_LISTEN`**
  S16 production-control-room TSR endpoint.
- **`CIVICCAST_PORTAL_BASE`**
  Resident portal URL handles.
- **`CIVICCAST_PORTAL_TOKEN`**
  Resident portal URL handles.
- **`CIVICCAST_RESIDENT_PORTAL_URL`**
  Resident portal URL handles.

AI model dispatch

- **`CIVICCAST_OLLAMA_BASE_URL`** — Local Ollama endpoint for the on-station model tier.
- **`CIVICCAST_PROVIDER_API_KEY`** — Cloud-provider key for Ollama Cloud or OpenRouter. (Prefer the keyring path: `civiccast model set-provider-key`.)

Diagnostics and overrides

- **`CIVICCAST_OPERATOR_CONSOLE_DIST`** — Override the prebuilt frontend bundles.
- **`CIVICCAST_OPERATOR_CONSOLE_URL`** — Override the prebuilt frontend bundles.
- **`CIVICCAST_PUBLIC_PORTAL_DIST`** — Override the prebuilt frontend bundles.
- **`CIVICCAST_APP_BUILD_ARTIFACTS_DIR`** — OTT app-build orchestrator paths.
- **`CIVICCAST_APP_BUILD_STORE_PATH`** — OTT app-build orchestrator paths.
- **`CIVICCAST_APP_PLATFORM_CONFIG_PATH`** — OTT app-build orchestrator paths.
- **`CIVICCAST_CERT_ROOT`** — Local-CA service-cert root override.
- **`CIVICCAST_REQUIRE_FCC_ACK`** — Require an FCC-rule acknowledgement before EAS surfaces.
- **`CIVICCAST_ROLLBACK_ARTIFACT_PATH`** — Path to a rollback artifact.
- **`CIVICCAST_TESTER_AVAILABLE_VERSION`** — Tester-rig overrides; rarely used in production.
- **`CIVICCAST_TESTER_OPS_STATE_PATH`** — Tester-rig overrides; rarely used in production.
- **`CIVICCAST_PLAYBACK_POLICY_STATE_PATH`** — Tester-rig overrides; rarely used in production.
- **`CIVICCAST_STATION_STATE_PATH`** — Tester-rig overrides; rarely used in production.
- **`CIVICCAST_SUPPORT_BUNDLE_DIR`** — Tester-rig overrides; rarely used in production.
- **`CIVICCAST_RUN_POSTGRES_TESTS`** — Run Postgres-only test suites against an external server.

Credential Store

Three families of credentials are kept in a dedicated store rather than environment variables so a station can rotate them without restarting the service.

Subsystem	Credential	Stored As	Configuration
Alerting (Twilio SMS)	account SID, auth token, sender number	JSON file referenced by `CIVIC-CAST_ALERT_CREDENTIALS_FILE`, with file-mode enforcement on POSIX hosts	Set the path env var, write the JSON, restart.
Subscription paywall (Stripe)	publishable key, secret key, webhook signing secret	Paywall service secret store; never written to logs	Configured in the operator console under *Settings → Paywall*. Default off (S26, V1.x).
RFC 3161 TSA	endpoint URL, optional policy OID	`CIVICCAST_TSA_URL` and `CIVIC-CAST_TSA_POLICY_OID`	FreeTSA is the default; bring your own TSA in production.
Cloud AI providers	per-provider API key	OS keyring, via `civiccast model set-provider-key`	Keys are write-only — never echoed back.

The store never logs a credential. The most an operator sees is a *stored / not-stored* line and a credential reference label.

CLI Reference

`civiccast` is a Typer-based CLI. Most operators rarely need it — the operator console is the day-to-day surface — but integrators and on-call use it heavily.

```
civiccast --version
civiccast doctor [--json] [--disk PATH]

civiccast installer plan \
    [--profile public-meetings] [--recommended-tier tier-1]
civiccast installer health-check [--profile public-meetings]
civiccast installer platform-plan [--os-family linux|macos]
    # Generic Linux/macOS bootstrap planning only. Windows deployment
    # readiness is decided separately by the native station's own
    # activation state (civiccast.installer.service), not this plan.
civiccast installer verify-package --artifact PATH --sidecar PATH
civiccast installer summary
civiccast installer beta-handoff \
    [--release-manifest PATH] [--clean-windows-evidence PATH]

civiccast model download [--offline-bundle --bundle-dir PATH] \
    [--profile NAME] [--dry-run]
civiccast model state
civiccast model import-offline --bundle-dir PATH \
    --expected file=sha256 [--expected ...]
civiccast model set-provider-key ollama-cloud|openrouter \
```

```
    [--key SECRET] [--clear]

civiccast cert rotate IDENTITY

civiccast token …               (managed by the operator console; see /api/tokens)

civiccast cable package --asset-id ID --title T --media F \
    --captions F --output-dir D
civiccast cable ndi-check
civiccast cable ndi-plan --media F --ndi-name N \
    [--muxer libndi_newtek]

civiccast activitypub …         (key + config helpers; see civiccast/activitypub/)

civiccast egress run --channel-id CH [--work-dir D] \
    [--poll-seconds 2] [--once]
civiccast egress verify --channel-id CH [--seconds 10]
civiccast egress recovery-proof --channel-id CH \
    --measured-seconds N …
civiccast egress continuity-proof --source-plan-json J \
    --config-json J \
    --output-path F
civiccast egress srt-continuity-proof \
    --source-plan-json J --config-json J \
    --sender-url U --receiver-url U \
    --receiver-output-path F
civiccast egress caption-decode-proof --channel-id CH …
civiccast egress trim-health [--older-than-days 30] [--dry-run]

civiccast live-takeover …       (cut a channel to a live source and return it)
```

Every subcommand supports `--json` for machine-readable output. Long-running ones (`civiccast egress run`) respond to SIGTERM and SIGINT cleanly.

Headless callers should authenticate via `CIVICCAST_STAFF_TOKENS` rather than passing tokens on the command line. Recovery codes, provider keys, and Stripe secrets must never appear on a command line in production.

Troubleshooting Matrix

Symptom	First Check	Then
Operator console shows **Do not broadcast yet**.	Open System Health. Read each not-ready row.	The line points to the next action (Setup → Storage, Setup → Credentials, etc.).

Symptom	First Check	Then
Live stream visible on operator console, blank on the public website.	`CIVICCAST_PUBLIC_BASE_URL` matches what residents actually load.	Confirm `CIVICCAST_CDN_PROVIDER` and `CIVICCAST_TRUSTED_PROXY_CIDRS` if the station is behind BunnyCDN / Cloudflare.
Captions empty or stuck.	`civiccast doctor` — confirm CPU/RAM tier; `CIVICCAST_OLLAMA_BASE_URL` reachable.	Look in the caption-tap directory (`CIVICCAST_CAPTION_TAP_DIR`) for stale segments.
Recording missing after a meeting.	`civiccast egress run --once --channel-id CH` for a snapshot of the egress worker state.	Check the recording finalizer worker (`CIVICCAST_FINALIZATION_WORKER`) and its logs.
Twilio SMS alerts not firing.	`CIVICCAST_ALERTING=true`; `CIVICCAST_ALERT_CREDENTIALS_FILE` points at a readable JSON.	Run an alert self-test from the operator console (System Health → Alerting).
EAS surface stuck on **needs FCC ack**.	Confirm the FCC acknowledgement is enabled and recorded in setup.	Re-open the EAS settings screen and confirm.
Trim re-render never finishes.	Recording finalizer worker is running.	Inspect the `civiccast/live/finalization_worker` logs for the asset id.
Behind a CDN, source-IP filters block legitimate residents.	`CIVICCAST_CDN_PROVIDER` is set and the preset is current.	Add explicit CIDRs in `CIVICCAST_TRUSTED_PROXY_CIDRS`.
`civiccast doctor` reports a tier below what the box should provide.	`--disk` points at the durable-storage volume.	Re-run with the explicit `--disk` argument.
Cloud AI key looks set but dispatch is offline.	`civiccast model set-provider-key PROVIDER` — confirm *stored*.	Re-check that the model the operator picked is supported by the saved provider.

For deeper trouble — TSDuck failures, GStreamer decode-back proofs, veraPDF validation, certificate rotation, mTLS — see Technical Operations Reference.

The Windows installer provisions the native beta runtime directly on the host, under the bundled runtime tree at `C:\Program Files\CivicCast (Native)\runtime\`: Python, FFmpeg/FFprobe, the CivicCast-bundled private GStreamer runtime with the caption-SEI elements, and Faster Whisper. TSDuck for cable verification is fetched and verified on demand into a contained per-user directory when cable verification is enabled — no admin rights, no system installer. NDI runtime/SDK, DeckLink hardware/drivers, app-store provider accounts, and live station headend equipment remain operator/provider supplied.

Local AI provisioning. CivicCast also provisions the local Ollama runtime for on-station AI (reusing a healthy existing install if one is already present and installing a pinned version only when Ollama is absent), then ensures the same three-tag target set of standard summary and translation models, downloading only the tags still missing, in the background after the operator console is already reachable; a model-download failure is reported honestly without blocking the rest of the install.

Cross-references

- Admin Guide — first install, recovery, backups.
- Meeting Operator Guide — night-of-meeting.
- Records Clerk Guide — caption review, publish.
- Technical Operations Reference — exact commands, certificate operations, release proofs.
- Operator Language Guide — UI vocabulary.
- Channel Egress Operator And Tester Runbook.
- Windows Release Trust And Verification.

Section C — Architecture Reference

This section is for developers, integrators, and external auditors who need to understand the source architecture and its relationship to the CivicCast 3.0 station-in-a-box specification. Every claim below cross-references a spec section or a concrete module path.

The canonical specification is at docs/spec/3.0/civiccast-3.0-station-in-a-box-MASTER.md. The repo-verified per-section status manifest is at docs/spec/3.0/ROADMAP.status.yaml. The migration-chain reconciliation history is at docs/spec/3.0/RECONCILIATION.md.

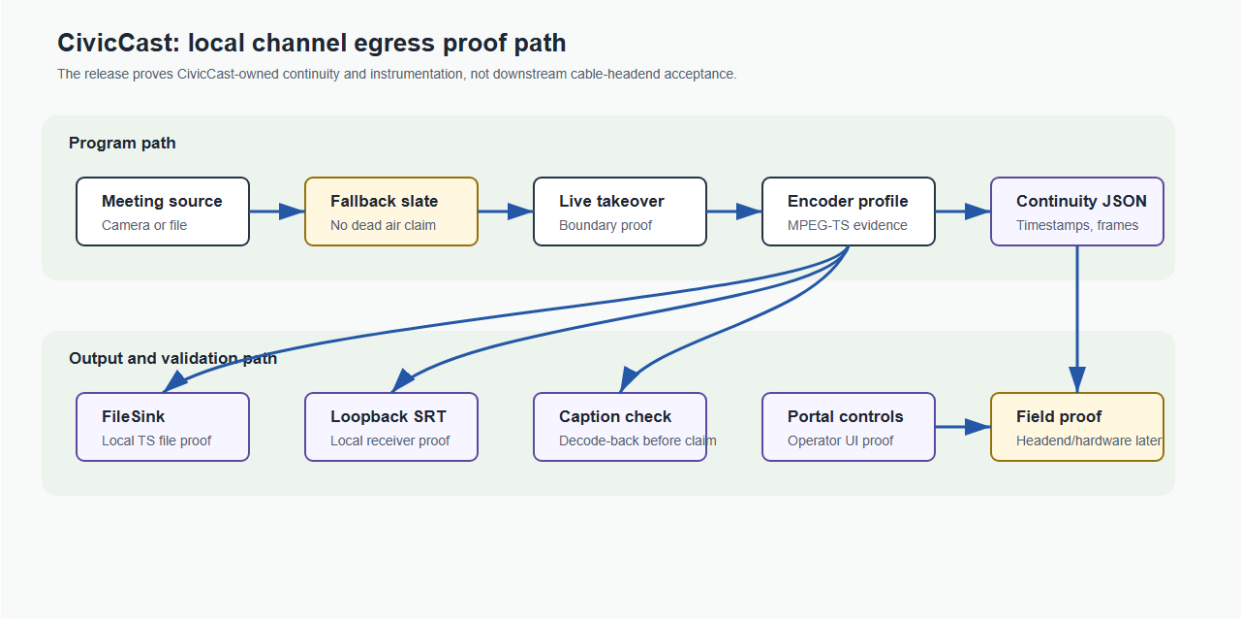


Figure 2: CivicCast channel-egress architecture

Subsystems And Module Layout

```
civiccast/
├── app.py                    FastAPI umbrella
├── cli.py                    Typer CLI (Section B)
├── auth/                     Identity, roles, staff tokens
├── installer/                Commissioning wizard (S3), first-boot
├── egress/                   Channel egress; gst/ = S15 engine
│   ├── gst/engine.py             GstPlayoutEngine
│   └── service.py                Egress daemon + supervisor (S9)
├── facility/                 Commit-to-Air (S4) + force matrix (S5)
├── cg/                       CG bulletin board (S6)
├── schedule/                 Programlog, autoschedule (S7, S19)
├── live/                     Live meeting + finalization (S7 finalizer)
├── recording/                Scheduled recording (S21)
├── metadata/                 Custom fields (S22)
├── reporting/                As-run + EPG + franchise (S23)
├── underwriting/             Spot management + affidavits (S24)
├── agenda/                   Meeting agenda + chapters (S25)
├── paywall/                  Subscription paywall (S26, default off)
├── alerting/                 Health + Twilio SMS (S8)
├── eas/                      Public-safety surface (S11)
├── captions/                 Live caption tap (S11)
├── records/                  Signed PDF + RFC 3161 (S7 + records lane)
├── analytics/                Resident-portal beacons + audience reports (S14)
├── ai_models/                Operator-chosen model dispatch (S13)
├── app_platform/             OTT build orchestration (S12)
├── apps/ott-native/          Native source for 6 OTT targets (S12 D1 closed)
├── control_room/             TSR over OBS/vMix/ATEM (S16, optional)
├── live/contribution/        VDO.Ninja remote guests (S17, optional)
├── archive/                  Internet Archive + local NAS
├── syndicate/                YouTube
├── subscribe/                Email + webhook subscribers
├── activitypub/              ActivityPub federation
├── stream/cdn/               BunnyCDN / Cloudflare R2 origin
└── common/trusted_proxy.py   CDN-aware client-IP resolver
```

The Five-Role Model

Defined in `civiccast/auth/roles.py`:

```
KNOWN_ROLES = (
    "setup_admin",
    "meeting_operator",
    "records_clerk",
    "publish_operator",
```

```
    "support_admin",
)
```

The authoritative permission matrix is the per-endpoint dependency in each router (`require_role("…")`). A staff member can hold multiple roles; the OR of their roles is what the dependency checks. Roles are stable across releases; new roles require a migration and a deprecation window.

The Alembic Migration Chain

The migration chain is single-headed at `0072_normalize_recording_file_uris`. The shape — including the late-landing sibling for S21 — is below in ASCII form; the canonical version (with each revision's down-revision pointer and the reasoning for the sibling) is in RECONCILIATION.md.

```
0049_per_sink_loudness
  -> 0050_caption_proof_samples
  -> 0051_public_safety_eas
  -> 0052_secondary_audio
  -> 0053_ai_model_configuration
  -> 0054_custom_metadata_fields (S22)
  -> 0055_asrun_and_epg (S23)
     |-- 0056_scheduled_recording (S21) --|
     `-- 0057_underwriting_spots (S24)    |
          -> 0058_meeting_agenda (S25)    |
          -> 0059_paywall_access (S26)    |
                    0060_recording_paywall_merge
  -> 0061_control_room_mode_gate
  -> 0062_media_integrity_columns
  -> 0063_producer_ops
  -> 0064_control_room_health_and_versioning
  -> 0065_recording_dropout_fields
  -> 0066_hls_sink_kind
  -> 0067_agenda_import_provenance
  -> 0068_migrate_batches
  -> 0069_control_room_session_surface_lock
  -> 0070_grandfather_scheduled_to_published
  -> 0071_published_blocks_overlap
  -> 0072_normalize_recording_file_uris (HEAD)
```

`0060` is the historical data-free merge revision that unified the `0056` sibling with the `0059` linear head. Later subsystem revisions extend that line through the current `0071` head. Automated schema-head tests enforce that single-head identity; ROADMAP.status.yaml tracks the capability evidence associated with the chain.

The S15 GStreamer Playout Engine

The S15 playout engine (`civiccast/egress/gst/`) replaces the previous ffmpeg-relay model with a persistent GStreamer pipeline plus hot-swap. The two relevant pieces:

- `engine.py` — the `GstPlayoutEngine` itself. A persistent pipeline configured around `GstInterpipe` / `input-selector` for seamless source swap (S15, also master §10 step 0 + step 1).
- `graph.py` — pipeline graph definitions; the playout, file-sink, SRT, and NDI shapes used by the

egress daemon.
- `worker.py` — the in-process worker the egress daemon supervises. This is what the beta release-artifact soak (step 13) targets.

`gstreamer` is the default engine, matching the v3.0 station-in-a-box specification's recommendation and the native station bootstrap's own runtime contract. Set `CIVICCAST_EGRESS_ENGINE=ffmpeg-concat` to opt back into the legacy ffmpeg-relay model for backwards compatibility.

The Three Protocol Seams

The three modules built last in the v3.0 sprint (scheduled recording, asset finalization, alerting) connect to the rest of the station via explicit `typing.Protocol` seams. The pattern lives in `civiccast/recording/service.py`, which is the most thorough example.

```
class CapturePipelineProtocol(Protocol):
    """Start, stop, and observe a capture pipeline for a scheduled recording."""

class AssetFinalizerProtocol(Protocol):
    """Hand a finished capture to the publish pipeline as a registered Asset."""

class AlertSinkProtocol(Protocol):
    """Receive scheduled-recording health / failure events as operator alerts."""
```

`RecordingService.__init__` takes each Protocol as an optional parameter. When `None`, the feature is treated as disabled (no capture runs, no asset finalization, no alerts), which keeps the data layer honest. The production FastAPI factory now binds:

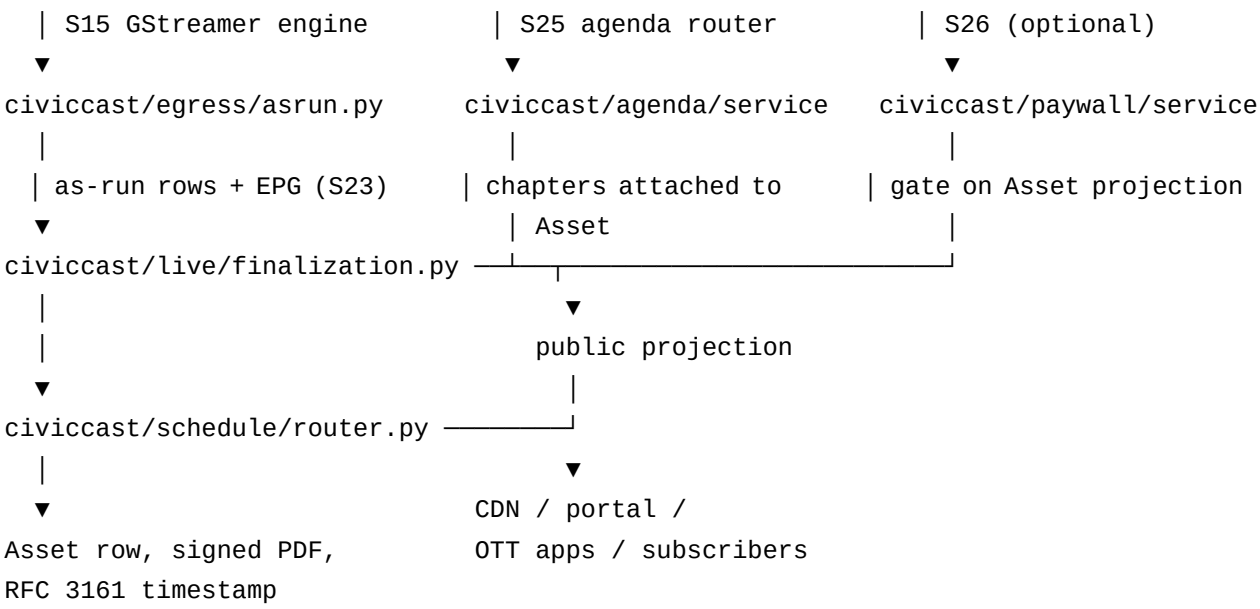
- `CapturePipelineProtocol` -> `FfmpegScheduledCapturePipeline`, which captures SDI/HDMI/NDI and network streams through the FFmpeg runtime.
- `RecordingInputPresetCatalog` discovers DeckLink SDI and Windows DirectShow HDMI capture devices and turns the operator's stable preset selection into exact FFmpeg input arguments. The Scheduled Recording screen shows detected or configured inputs; if none are available, it says so and will not accept a made-up device name. Use `CIVICCAST_RECORDING_INPUT_PRESETS_JSON` for a connector label, DeckLink format code, or paired DirectShow audio device that discovery cannot infer.
- `AssetFinalizerProtocol` -> `ScheduledRecordingAssetFinalizer`, which probes the recorded file, validates ingest metadata, and registers a recorded `Asset`.
- `AlertSinkProtocol` -> `RecordingAlertSink`, which sends source and finalization failures to the S8 condition hub.

This shape is what lets S21 ship as a complete service + API + UI in its own slice without entangling the egress, capture, and alerting subsystems. Each Protocol is independently mockable and independently testable.

The Publish Pipeline

Recordings, meeting agendas, and chapter markers tie into the public Asset graph through the S7 media-lifecycle subsystem. The relevant flow:

```
[capture]                    [meeting agenda]          [paywall]
```



Finalization and packaging create the private media package. Public routes also require `published_at`, which is set only after the Portal surface succeeds in the publish workflow. An unpublished package returns 404 from both public metadata and `/media/vod`; packaging is never treated as approval.

The `schedule_adapter`/`asrun_recorder` in `civiccast/reporting/` record what actually went on air, producing the as-run reports the franchise authority sees. EPG export (`civiccast/reporting/epg.py`) emits XMLTV for downstream guides.

The CDN-Aware Trusted-Proxy Resolver

Located at `civiccast/common/trusted_proxy.py`. Everything that consumes "real client IP" goes through this resolver: paywall rate limiter, recording cross-station guard, audit logger, analytics ingestion filter.

```
incoming request
      |
      ▼
request.client.host = X
      |
      ▼
walk X-Forwarded-For right-to-left
      |
      ├── if X is a trusted proxy → trust the chain
      |     trusted = CIVICCAST_TRUSTED_PROXY_CIDRS
      |             + provider preset (CIVICCAST_CDN_PROVIDER)
      |             + private/loopback (CIVICCAST_TRUST_PRIVATE_PROXIES)
      |
      └── else → the immediate peer IS the real client
                 (X-Forwarded-For is ignored — anti-spoof)
```

Provider presets ship for `bunny` and `cloudflare` / `cloudflare_r2`, snapshotted from the providers' pub-

lished edge-CIDR lists. Operators should reconcile the snapshot with the provider's current documentation at deploy time; CivicCast does not auto-update at runtime.

Comparative Capability Status

The repository software-capability table is tracked in ROADMAP.status.yaml. A `shipped` row means the software surface exists; it does not by itself prove the installer, station hardware, external provider, or field workflow:

PEG automation surface	CivicCast section	Status
Playout engine	S15	shipped (GStreamer)
Commit-to-Air / operator takeover controls	S4 / S5	shipped
CG / bulletin board	S6	shipped
Health + alerting	S8	shipped (Twilio SMS adapter is real)
OTT apps	S12	shipped (6 native source trees)
AI model selection	S13	shipped
Captions / Loudness / EAS	S11	shipped
Scheduling automation	S19	shipped
Production control room	S16	shipped (optional tier)
Remote contribution	S17	shipped (optional tier)
Scheduled recording	S21	shipped
Custom metadata fields	S22	shipped
As-run / EPG / franchise	S23	shipped
Underwriting spots + affidavits	S24	shipped
Meeting agenda + chapters	S25	shipped
Subscription paywall	S26	backend/operator surface built; public tier discovery is not deployed, so the flow is not end-to-end usable
Analytics / audience measurement	S14	built — packaged audience reports (CSV/XML export); durable Postgres store + dashboard UI is a tracked follow-on
Accessibility WCAG 2.1 AA	S20	built — axe-core + DC-4 contrast gate (both portals); screen-reader walkthrough is manual/beta

Hardware-bounded surfaces — physical SDI, QAM modulation, EAS hardware endec, app-store publication — are out of the software boundary; the software writes the full path up to the interface, and the LPM-lab acceptance (Step 14) is where the hardware side gets proven.

Read every `shipped` row above as "the source module exists and has unit/ integration coverage," not as "field-proven for this beta." The Advanced Capabilities — Roadmap, Not This Beta box in Section A states, in plain language, which of these a station should not yet rely on: full cable/SDI headend delivery, simultaneous multi-channel operation, turnkey Internet Archive/YouTube syndication, OTT app-store publication, and any EAS row — CivicCast is EAS-adjacent software, not a certified EAS device.

Open Items And Roadmap Pointers

One external release gate remains before CivicCast can be described as a managed-service-quality claim:

- **Step 13 — release-artifact soak.** Earlier soak evidence remains historical input. A release cannot inherit exact-artifact proof from an earlier candidate; its own signed public executable must complete clean-host installation, lifecycle proof, publication, and public-download verification.
- **Step 14 — Headend acceptance at the LPM lab.** First-station beta at the LPM headend; external evidence only.

S14 and S20 closed their pinned gaps this pass, with two honestly-scoped follow-ons still open:

- **S14 — Audience reports.** `civiccast/analytics/audience_reports.py` ships franchise audience reports (per-channel totals, top assets, CSV/ XML export) over the existing aggregate analytics store. Still open, tracked separately: the durable Postgres-backed event/rollup tables (the store is still a single JSON file), the four-panel dashboard UI, a board-ready PDF, and year-over-year trend.
- **S20 — Accessibility.** axe-core CI gates plus a real DC-4 contrast gate (both portals, with an empty-scan honesty guard and a permanent falsification test) are in place. Lighthouse's accessibility category is deliberately not run as a separate CI step — it audits via axe-core internally, so the existing zero-violation axe gate is already a stricter floor. The screen-reader (NVDA/VoiceOver) walkthrough remains a manual, beta-stage item for the master §12 release `/walkthrough`.

The remaining §10/§11 source modules are present, but module presence is not field proof. The stock acceptance claim remains limited to the installer and the recorded-sample rehearsal → private package → Portal approval → resident playback path described in Section A. Live ingest, 24/7 operation, station devices, and external providers require separate proof.

For deeper section reading:

- S15 — Playout engine (GStreamer)
- S7 — Media lifecycle & readiness
- S8 — Health, alerting, support updates
- Legal Notices
- Patent Risk Notes
- S21 — Scheduled recording
- S25 — Meeting-agenda integration
- S26 — Subscription paywall
- S10 — Field certification & proof ladder

Language Standard

All user-facing CivicCast copy follows the Operator Language Guide. In short:

- **Broadcast** for the live or recorded meeting the public watches.
- **Publish** for making approved records available after review.
- **Not set up yet** or **needs IT help** instead of *blocked* when the next step is configuration.
- **Ready**, **check before meeting**, or **do not broadcast yet** instead of internal health-state names.
- **Auto-generated captions** and **reviewed captions** so the difference is clear to staff and to residents.